Telemetry Anomaly Monitor

Technical Report — OPS-SAT Telemetry Anomaly Detection

1. Executive Summary

This project converts the supplied telemetry analysis into a deployable anomaly-detection system. It uses the dataset's official train/test boundary, selects an operating threshold on a validation subset of training data, trains an ExtraTrees classifier, and exposes the same artifacts through Streamlit and Flask. An Isolation Forest is also included as a complementary unlabeled detector.

2. Dataset

The segment dataset contains 2,123 rows, 434 anomalies, 1,689 nominal segments and 9 telemetry channels. The companion signal table contains 303,493 observations. Segment-level features include distribution statistics, peaks, derivatives, duration, length, gap measures and normalized variance measures.

3. Model

The primary model is ExtraTreesClassifier with 500 trees, minimum leaf size 2, balanced class weights and random seed 42. Numeric features are median-imputed; channel is most-frequent-imputed and one-hot encoded. The threshold was selected on a stratified validation subset by maximizing F1.

4. Evaluation

The official holdout contains 529 rows. Results: accuracy 0.9735, precision 0.9541, recall 0.9204, F1 0.9369, ROC-AUC 0.9917 and PR-AUC 0.9780. Confusion matrix: 411 true negatives, 5 false positives, 9 false negatives and 104 true positives. Threshold = 0.545.

5. Streamlit

The Streamlit application provides Dashboard, Single Segment and Batch CSV modes. It displays anomaly probabilities, channel-level anomaly rates, a feature-space plot and downloadable batch predictions.

6. Flask

The Flask application provides a browser form, GET /health and POST /predict JSON inference. Both interfaces load the same saved ExtraTrees artifact, ensuring consistent inference behavior.

7. Limitations

These results describe this supplied dataset, not production spacecraft performance. The segment dataset is modest in size and some channels have very few examples. Time-aware validation, drift detection, calibration, authentication, logging and operational alerting should be added before production use.

8. Future Work

Promising upgrades include raw-signal CNN/LSTM/Transformer models, channel-specific baselines, hybrid supervised/unsupervised scoring, feature explanations, streaming ingestion, alert history and automated drift-triggered retraining.

Key Metrics

Metric	Value
Accuracy	0.9735
Precision	0.9541
Recall	0.9204
F1	0.9369
ROC-AUC	0.9917
PR-AUC	0.9780
Threshold	0.545